

# DSA 8070: Introduction and Multivariate Data Exploration

From multiple variables to multivariate structure

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Reading: Zelterman (2015), Chapters 1 and 3

# Agenda

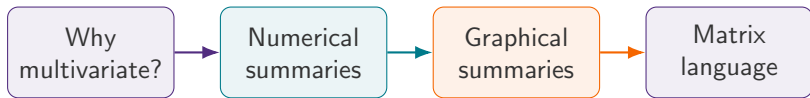
Course Overview

Numerical Exploration

Graphical Exploration

Summary

# Roadmap for This Lecture



## Key idea

Multivariate analysis is about learning from variables jointly – especially how they vary together.

# What Makes Data Multivariate?

## Univariate

One measurement per observational unit:

$$X_i, \quad i = 1, \dots, n.$$

## Multivariate

Several measurements on the same unit:

$$\mathbf{X}_i = \begin{bmatrix} X_{i1} \\ \vdots \\ X_{ip} \end{bmatrix}.$$

```
> head(Boston)
```

|   | crim    | zn | indus | chas | nox   | rm    | age  | dis    | rad | tax | ptratio | black  | lstat | medv |
|---|---------|----|-------|------|-------|-------|------|--------|-----|-----|---------|--------|-------|------|
| 1 | 0.00632 | 18 | 2.31  | 0    | 0.538 | 6.575 | 65.2 | 4.0900 | 1   | 296 | 15.3    | 396.90 | 4.98  | 24.0 |
| 2 | 0.02731 | 0  | 7.07  | 0    | 0.469 | 6.421 | 78.9 | 4.9671 | 2   | 242 | 17.8    | 396.90 | 9.14  | 21.6 |
| 3 | 0.02729 | 0  | 7.07  | 0    | 0.469 | 7.185 | 61.1 | 4.9671 | 2   | 242 | 17.8    | 392.83 | 4.03  | 34.7 |
| 4 | 0.03237 | 0  | 2.18  | 0    | 0.458 | 6.998 | 45.8 | 6.0622 | 3   | 222 | 18.7    | 394.63 | 2.94  | 33.4 |
| 5 | 0.06905 | 0  | 2.18  | 0    | 0.458 | 7.147 | 54.2 | 6.0622 | 3   | 222 | 18.7    | 396.90 | 5.33  | 36.2 |
| 6 | 0.02985 | 0  | 2.18  | 0    | 0.458 | 6.430 | 58.7 | 6.0622 | 3   | 222 | 18.7    | 394.12 | 5.21  | 28.7 |

Multivariate analysis consists of statistical methods for analyzing these measurements jointly.

# Why Univariate Analysis Can Miss the Point

- ▶ Separate summaries describe each variable one at a time.
- ▶ They do *not* describe how variables move together.
- ▶ Association among variables can reveal structure that is invisible marginally.

## A useful question

If two variables are strongly related, what information is lost when we analyze them completely separately?

## Key idea

Multivariate methods exploit dependence rather than treating it as a nuisance.

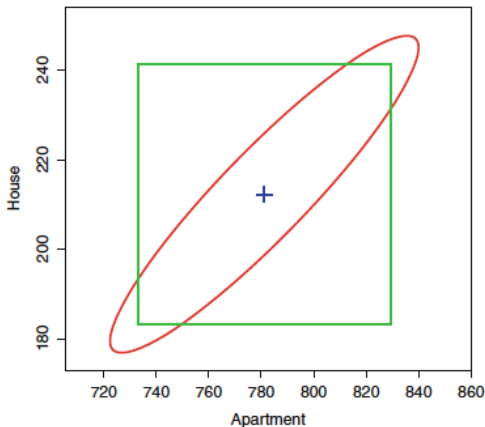
# Why Multivariate Analysis Matters in Modern Data Science

| Multivariate statistics    | Data science / ML                   |
|----------------------------|-------------------------------------|
| Dependence among variables | Feature relationships               |
| Dimension reduction        | Feature extraction / representation |
| Classification             | Supervised learning                 |
| Clustering                 | Unsupervised learning               |
| Statistical distance       | Similarity / anomaly detection      |
| Multivariate prediction    | Multi-output prediction             |

## Key idea

Many modern machine-learning tasks are fundamentally multivariate problems.

# Joint Analysis Can Lead to Sharper Inference



## Interpretation

When variables are related, the joint covariance structure changes the geometry of uncertainty. A multivariate analysis can therefore answer questions that separate univariate intervals cannot.

## Four Recurring Multivariate Questions

1. **Reduce dimension:** Can many variables be summarized by a few directions or scores?
2. **Group or classify:** Which observations are similar, and to which group does a new observation belong?
3. **Relate and predict:** How are sets of variables associated, and can some predict others?
4. **Compare groups:** Do populations differ in their mean vectors or covariance structures?

Where this appears later

PCA, factor analysis, clustering, classification, multivariate regression, canonical correlation, Hotelling's  $T^2$ , and MANOVA.

# Dimension Reduction or Structural Simplification

- ▶ **Goal:** replace many correlated variables with a smaller number of informative summaries.
- ▶ A useful summary should retain important variation while reducing complexity.

$$(X_1, \dots, X_p) \longrightarrow Z_1, \dots, Z_k, \quad k \ll p.$$

## Examples later in the course

Principal component analysis, factor analysis, multidimensional scaling, and nonlinear embedding.

# Grouping and Classification

## Clustering

Groups are *not* known in advance.

data  $\longrightarrow$  discover groups

## Classification

Groups are already defined.

training data  $\longrightarrow$  classify new units

Both require a notion of similarity or distance in a multivariate space.

# Dependence and Prediction

- ▶ Estimate relationships among several variables.
- ▶ Predict one set of variables from another.
- ▶ Study association between two multivariate sets.

$$\underbrace{(X_1, \dots, X_p)}_{\text{predictors}} \longrightarrow \underbrace{(Y_1, \dots, Y_q)}_{\text{responses}}$$

Where this appears later

Multivariate regression and canonical correlation analysis.

# Hypothesis Testing Becomes Multivariate

Instead of comparing one mean, we may compare an entire mean vector:

$$H_0 : \mu_1 = \mu_2.$$

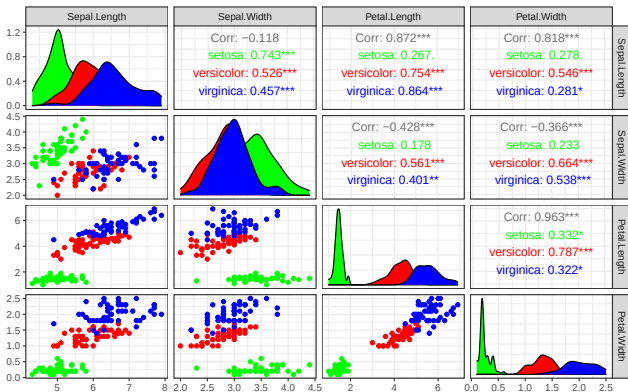
Or we may ask whether covariance structures differ:

$$H_0 : \Sigma_1 = \Sigma_2.$$

## Why this matters

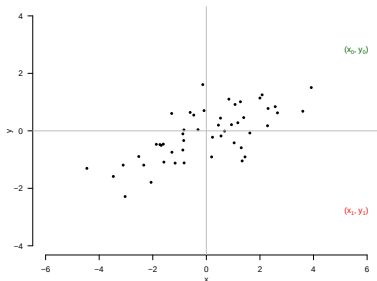
Testing variables one-by-one can inflate multiplicity and ignore covariance among outcomes. Multivariate tests address the joint question directly.

# Exploratory Data Analysis Is Still the First Step



Before choosing a formal model, look for association, group separation, unusual observations, skewness, and scale differences.

# Statistical Distance: “Close” Depends on Covariance



- ▶ Euclidean distance treats every direction equally.
- ▶ Multivariate data often have different scales and correlated directions.
- ▶ A covariance-aware distance can better reflect what is statistically unusual.

Where this appears later

Mahalanobis distance in the matrix-algebra lecture.

# A Preview of the Mathematical Language

$$X_{n \times p} = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{bmatrix}.$$

Location

$$\hat{\mu} = \bar{x}$$

Variation and dependence

$$\hat{\Sigma} = S$$

Eigenvalues and eigenvectors then describe important directions and magnitudes of variation.

# Multivariate Normal Distribution: A Preview

For a  $p$ -dimensional random vector,

$$\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma),$$

the density is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}.$$

- ▶  $\boldsymbol{\mu}$  determines the center.
- ▶  $\Sigma$  determines scale, correlation, and orientation.
- ▶ The quadratic form is the squared Mahalanobis distance.

## Where this appears later

Week 3: multivariate normal models, copulas, and alternatives to normality.

# Organizing Multivariate Data

For  $n$  observational units and  $p$  variables,

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}.$$

- ▶  $x_{ik}$ : variable  $k$  measured on unit  $i$ .
- ▶ Rows correspond to observations.
- ▶ Columns correspond to variables.

## Quick check

If  $X$  has dimension  $150 \times 4$ , what are  $n$  and  $p$ ?

# Sample Mean and Variance: One Variable at a Time

For variable  $k$ ,

$$\bar{x}_k = \frac{1}{n} \sum_{i=1}^n x_{ik}, \quad s_k^2 = \frac{1}{n-1} \sum_{i=1}^n (x_{ik} - \bar{x}_k)^2.$$

## Interpretation

$\bar{x}_k$  describes location and  $s_k^2$  describes marginal variation for variable  $k$ .

These are familiar univariate summaries. The multivariate step is to add summaries of how *pairs* of variables vary together.

# Sample Covariance: How Two Variables Vary Together

$$s_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

- ▶  $s_{jk} > 0$ : large values of one variable tend to accompany large values of the other.
- ▶  $s_{jk} < 0$ : large values of one tend to accompany small values of the other.
- ▶  $s_{jk} \approx 0$ : little *linear* co-movement.

```
## {r}
dat <- mvrnorm(n = 50, mu = c(0, 0), Sigma = matrix(c(1, 0, 0, 1), 2))
cov(dat[, 1], dat[, 2])
##
```

```
[1] -0.1508848
```

## Sample Correlation: Standardized Covariance

$$r_{jk} = \frac{s_{jk}}{\sqrt{s_{jj}}\sqrt{s_{kk}}}.$$

- ▶  $-1 \leq r_{jk} \leq 1$ .
- ▶  $r_{jk} = r_{kj}$ .
- ▶ Correlation removes measurement units, making associations easier to compare.
- ▶ Zero correlation does **not** imply independence in general.

### Key idea

Covariance carries units; correlation is scale-free. Both describe linear association.

## Collect the Summaries into Matrices

| Mean                                                                                | Covariance                            | Correlation                              |
|-------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------|
| $\bar{\mathbf{x}} = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{bmatrix}$ | $S = (s_{jk})_{p \times p}$ $S = S^T$ | $R = (r_{jk})_{p \times p}$ $r_{jj} = 1$ |

These objects become the basic input to many methods used later in the course.

# Generalized Variance: From Variance to Multivariate Volume

For a covariance matrix  $S$ ,

$$\text{generalized variance} = |S|.$$

```
```{r}
data(mtcars)
vars <- which(names(mtcars) %in% c("mpg", "disp", "hp", "drat", "wt"))
car <- mtcars[, vars]; S <- cov(car)
(genVar <- det(S))
```
```

```
[1] 3951786
```

## Preview

The determinant measures multivariate volume. It becomes small when variables are highly redundant and zero when the data collapse into a lower-dimensional subspace.

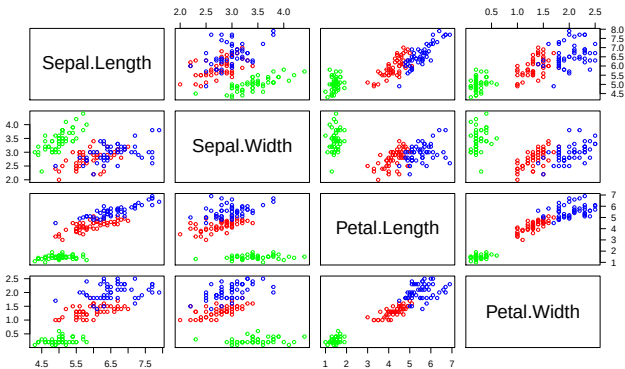
# Why Visualization Is Harder in High Dimensions

- ▶ For  $p = 1$ , we can use a histogram or boxplot.
- ▶ For  $p = 2$ , a scatterplot directly displays the joint structure.
- ▶ For  $p > 3$ , no single conventional plot can show all dimensions at once.

## Key idea

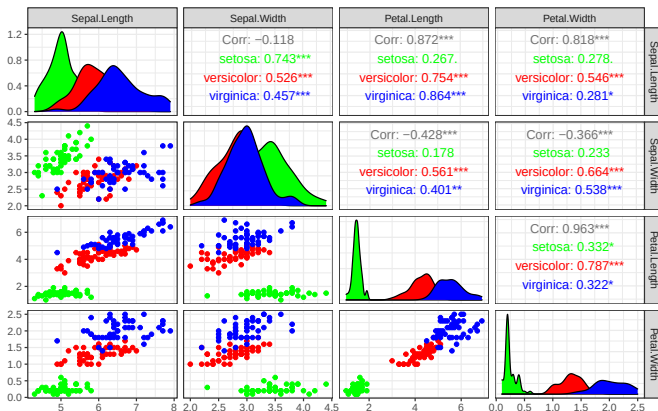
Multivariate visualization therefore uses coordinated views, projections, or summaries.

# Scatterplot Matrix: Start with Pairwise Relationships



Look for linear or nonlinear association, group separation, changing variability, and unusual observations.

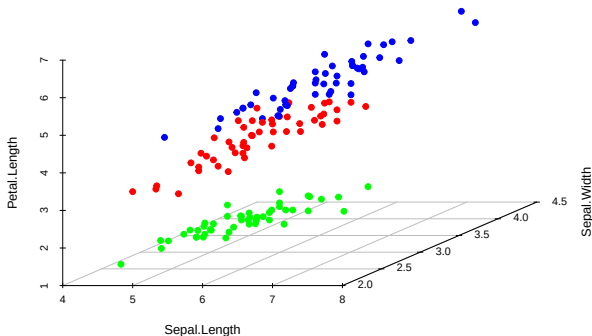
# ggpairs: Graphics + Numerical Summaries



## Why this is useful

A single display combines marginal distributions with pairwise dependence, giving a compact first look at a moderate number of variables.

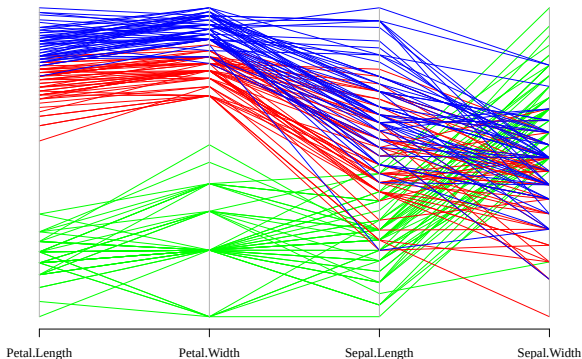
## 3D Scatterplots: Useful, but Limited



- ▶ Can reveal group structure across three variables.
- ▶ Rotation and perspective matter.
- ▶ Does not solve visualization once  $p$  becomes much larger than three.

## Parallel Coordinate Plots

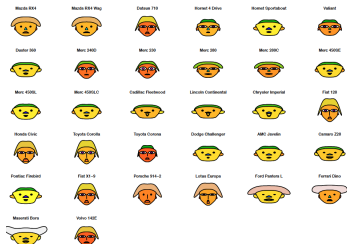
- ▶ Each observation becomes a line crossing one axis per variable.
- ▶ Useful for patterns, group differences, and unusual profiles.
- ▶ Scaling and variable order strongly affect interpretation.



# Chernoff Faces: A Historical Multivariate Display

```
> head(mtcars)
```

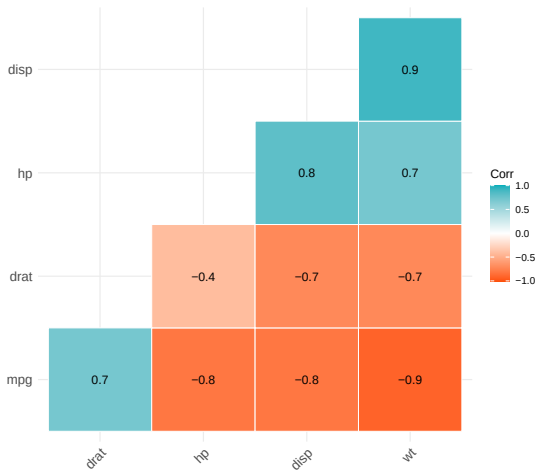
|                   | mpg  | cyl | displacement | horsepower | weight | quarter mile time | vs    | acceleration | gear | carburetors |
|-------------------|------|-----|--------------|------------|--------|-------------------|-------|--------------|------|-------------|
| Mazda RX4         | 21.0 | 6   | 160          | 110        | 3.50   | 2.620             | 16.46 | 0            | 1    | 4           |
| Mazda RX4 Wag     | 21.0 | 6   | 160          | 110        | 3.50   | 2.875             | 17.02 | 0            | 1    | 4           |
| Datsun 710        | 22.8 | 4   | 108          | 93         | 3.85   | 2.320             | 18.61 | 1            | 1    | 4           |
| Hornet 4 Drive    | 21.4 | 6   | 258          | 110        | 3.08   | 3.215             | 19.44 | 1            | 0    | 3           |
| Hornet Sportabout | 18.7 | 8   | 360          | 175        | 3.15   | 3.440             | 17.02 | 0            | 0    | 3           |
| Valiant           | 18.1 | 6   | 225          | 105        | 2.76   | 3.460             | 20.22 | 1            | 0    | 3           |



## Interpret with care

Chernoff faces illustrate the idea of mapping many variables to visual features, but perception of facial features is uneven. They are best viewed as a historical example rather than a default modern visualization.

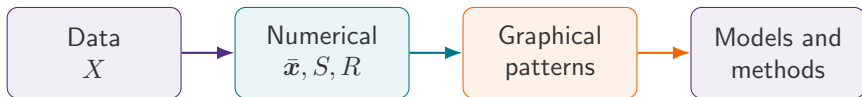
# Visualizing a Correlation Matrix



## Key idea

Visual summaries of  $R$  can reveal blocks of strongly related variables and potential redundancy.

# Putting the Pieces Together



- ▶ Multivariate analysis studies several variables jointly.
- ▶ Covariance and correlation summarize dependence.
- ▶ Numerical and graphical exploration should precede formal modeling.
- ▶ Matrix algebra provides the compact language connecting these ideas.

## Next: Matrix Algebra

$X \longrightarrow (\bar{\mathbf{x}}, S) \longrightarrow \text{geometry} \longrightarrow \text{multivariate methods}$

### Next lecture

We will review vectors, matrices, covariance geometry, eigenvalues/eigenvectors, quadratic forms, Mahalanobis distance, and other tools that recur throughout the course.